Carbon

Overview

SWAT+ exposes two soil carbon paths today. The choice is set by the integer `bsn_cc%cswat` in `codes.bsn`:

`cswat`	Model	Pools
0	Static (legacy)	Single residue + humus; C is not tracked as a state
1	C-FARM	Reserved for the C-FARM model; not implemented. Treated as off (no carbon model, no `carbon.bsn` required).
2	Active CENTURY/SWAT-C module	Three litter pools and three soil C pools (microbial, slow, passive) following Parton et al. (1993, 1994); driven by `cbn_zhang2.f90`

The carbon model controls how plant residue, manure, tillage, and grazing inputs are routed into the soil and the subsequent CO2, N and P fluxes.

> **Carbon activation value changed to 2**
>
> The dynamic carbon model now activates on `cswat = 2`, matching legacy SWAT numbering (1 = C-FARM, 2 = CENTURY). It was previously selected by `cswat = 1`. `carbon_bsn_read.f90` returns early for any value other than 2, so `cswat = 1` is now inert: no carbon model and no `carbon.bsn` required. This is an input-compatibility break: existing `codes.bsn` files (including refdata and the SWAT+ Editor default) left at `carbon = 1` will silently run with the carbon model off and must be changed to `carbon = 2`.

Process equations

Static (cswat = 0)

Mineralisation and immobilisation are computed in `nut_nminrl.f90`. There is no explicit soil C state. Plant residue carbon is implicit through the `0.42 * biomass` conversion. Residue decomposes at rate

```
decr = max(decr_min, min(1, rsdco_pl * ca * csf))
```

with 80% of mineralised N going to NO3 and 20% to active humus N. See nutrient cycling.

Active carbon module (cswat = 2)

Selected when `bsn_cc%cswat = 2`. The active path runs the full Century pool transformation in `cbn_zhang2.f90`, fed by residue inputs, microbial decomposition, and tillage-driven mixing. Basin-wide and per-layer coefficients come from `carbon.bsn` and `carbon_lyr.bsn`; both files are **required** when `cswat = 2` and are read by `carbon_bsn_read.f90`. The tillage method is set by `bsn_cc%idc_till`:

`idc_till`	Method
1	DSSAT-style
2	EPIC-style
3	Kemanian (default)

idc_till	Method
4	DNDC-style

In hru_control.f90, when cswat = 2:

```
call cbn_surfrsd_decomp
call cbn_rsd_transfer
call cbn_zhang2
```

Surface residue decomposes in cbn_surfrsd_decomp.f90 using the combined temperature-moisture factor csf from the N cycle. Below-ground residue is partitioned to soil pools through cbn_rsd_transfer.f90. The cbn_zhang2.f90 routine then runs the Century-style multi-pool turnover that C-FARM and Century share.

The active path also adds the plant biomass increment as carbon (hpc_d(j)%npp_c) and accumulates respiratory CO2.

Tillage residue mixing uses mgt_newtillmix_cswat1.f90 (vs mgt_newtillmix_cswat0.f90 for the static model). Tillage activity decays over till_eff_days days (a column of carbon.bsn), controlled per layer by soil(j)%ly(ly)%tillagef_tillmix.

Century pool structure (active path)

The Century scheme implemented in cbn_zhang2.f90. Pools:

- Litter: structural (ls), metabolic (lm), with the structural pool further split into lignin (lsl) and non-lignin
- Soil organic carbon: microbial biomass (bm), slow humus (hs), passive humus (hp)

For each pool, potential transformation under optimal conditions uses a first-order rate (the *_para rates in carbon_module.f90). Daily realised transformation is

```
rate(layer) = rate_optimal * cs
cs = sqrt(cdg * sut) * xbm * ox     (cs capped at 10)
```

cdg and sut are the temperature and water factors from the N cycle. ox is the oxygen factor

```
ox = 1 - OX_aa * depth / (depth + exp(OX_aa - OX_bb * depth))
```

with OX_aa = 10.0, OX_bb = 0.035. xbm is the texture and structure factor: 1 in the surface litter and 1 - 0.75 * (silt + clay) elsewhere.

Default Parton optimal rates (per day):

Pool	Surface	Subsurface
Microbial biomass (BMR)	0.0164	0.02
Slow humus (HSR)	-	0.000548
Passive humus (HPR)	-	0.000012
Metabolic litter (LMR)	0.0405	0.0507
Structural litter (LSR)	0.0107	0.0132

CO2 allocations on transformation:

From	To CO2 (surface)	To CO2 (subsurface)
Biomass	0.60	0.85 - 0.68 * (silt+clay)
Metabolic litter	0.60	0.55
Non-lignin structural	0.60	0.55
Lignin structural	0.30	0.30
Slow humus	0.55	0.55
Passive humus	0.55	0.55

Biomass-to-passive humus allocation in subsurface layers: `0.003 + 0.032 * clay`. Slow-to-passive allocation: `max(0.001, PRMT_45 - 0.00009 * clay)` with `PRMT_45 = 0.003`.

Microbial-biomass leaching uses a water-balance form

```
abl = 1 - exp(-vflow / (0.01 * st + KOC * bd))
```

with `KOC = 4000` (organic carbon partition coefficient, `part_DOC_para`).

Lignin shape factor for structural litter: `xlslf = exp(-3 * lslf)` where `lslf` is lignin fraction of structural litter.

C:N ratios are kept within Parton bounds for each pool and the transformation can release or consume mineral N depending on the deficit (`cpn1..cpn5` and `sum1..sum5`).

The Parton defaults are overridden by the per-layer rates and CO2 fractions supplied in `carbon_lyr.bsn`; the residue and basin-wide tunables come from `carbon.bsn`. The legacy `carb_coefs.cbn` and `basins_carbon.tes` files (and their readers `carbon_coef_read.f90` and `carbon_read.f90`) have been removed.

DOC and DIC

The Century implementation accumulates dissolved organic carbon (DOC) and dissolved inorganic carbon (DIC) percolation. Defaults `peroc_DIC_para = 0.95`, `peroc_DOC_para = 0.70`, `hlife_doc_para = 50` days for DOC half-life in groundwater. Source: `carbon_module.f90`, `cbn_zhang2.f90`.

Switches and parameters

File	Field	Default	Effect
`codes.bsn`	`cswat`	0	0 = static, 1 = C-FARM (reserved, inert), 2 = active CENTURY/SWAT-C module
`codes.bsn`	`idc_till`	3	Tillage method when `cswat = 2`
`carbon.bsn`	28 basin-wide scalars	see file	Required when `cswat = 2`; covers initial pool fractions, water coupling, manure partition, biomixing and tillage curves, temperature thresholds, residue C:N / C:P controls, and the slow-humus init method
`carbon_lyr.bsn`	per-layer rate constants and CO2 fractions	see file	Required when `cswat = 2`; one row per layer group
`carbon_layers.prt`	`cb_n_layers`	max HRU layers	Optional; fixes the number of layers in per-layer carbon output. If absent, defaults to the largest soil layer count across all HRUs

Notable Century coefficients (from `carbon_module.f90`):

Name	Default	Meaning
`CFB_para`	0.42	Carbon fraction of residue
`er_POC_para`	1.5	POC enrichment ratio
`peroc_DOC_para`	0.70	DOC percolation coefficient
`peroc_DIC_para`	0.95	DIC percolation coefficient
`part_DOC_para`	4000	DOC partition coefficient
`hlife_doc_para`	50 days	DOC half-life in groundwater
`ALSLCO2_para`	0.30	Lignin-to-CO2 allocation
`APCO2_para`	0.55	Passive humus to CO2
`ASCO2_para`	0.55	Slow humus to CO2
`BMR_para_sur` / `BMR_para_sub`	0.0164 / 0.02	Microbial biomass turnover (d^-1)
`HPR_para`	0.000012	Passive humus turnover (d^-1)
`HSR_para`	0.000548	Slow humus turnover (d^-1)
`PRMT_45_para`	0.003	Slow-to-passive allocation coefficient

Name	Default	Meaning
PRMT_51_para	1.0	Microbial activity coefficient for top layer
OX_aa_para	10.0	Oxygen factor coefficient
OX_bb_para	0.035	Oxygen factor coefficient

Implementation

Core: cbn_zhang2.f90 (Century pool transformations), cbn_surfrsd_decomp.f90 (surface residue), cbn_rsd_decomp.f90 (sub-surface residue, used in legacy paths), cbn_rsd_transfer.f90 (residue routing into Century pools).

Inputs: carbon_module.f90 (data types and module-scope tunables), carbon_bsn_read.f90 (reads both carbon.bsn and carbon_lyr.bsn), carbon_layers_read.f90 (optional carbon_layers.prt).

Tillage: mgt_newtillmix_cswat0.f90, mgt_newtillmix_cswat1.f90, mgt_newtillmix_wet.f90, mgt_tillfactor.f90, mgt_biomix.f90.

Driver in hru_control.f90 switches by bsn_cc%cswat. Outputs are written by output_landscape_init.f90, hru_carbon_output.f90, lsu_carbon_output.f90, and soil_nutcarb_write.f90. Output gating is per-family through the hru_cb_* and lsu_cb_* rows of print.prt; the old cbn_diagnostics flag has been retired.

Related pages

- codes.bsn. The cswat switch.
- carbon.bsn. Basin-wide carbon scalars.
- carbon_lyr.bsn. Per-layer carbon coefficients.
- Carbon outputs. HRU and LSU file families.
- Parameter change files. 40 wired carbon calibration parameters.
- Nutrient cycling.
- Plant growth.